

DeepScale: Modular Retrieval-Augmented Generation for Scientific Literature

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Abstract

1. Introduction

The primary objective of this paper is to resolve the acute hallucination problem in academic literature search and synthesize factual answers with verifiable page-level source citations. Traditional language models frequently confabulate experimental statistics or attribute claims to fictitious papers. We introduce DeepScale to bridge this gap by enforcing strict provenance checking and context-grounded generation. The overarching goal is to enable researchers to query complex multi-page preprints and immediately inspect the exact paragraphs, figures, and datasets that justify the model's generated conclusions.

2. Methodology

DeepScale employs a three-tiered architectural pipeline:

1. **Layout-Aware PDF Ingestion:** We utilize block-coordinate parsing to reconstruct reading flows across multi-column academic layouts, isolating section headers from page headers and footnotes.
2. **Section-Bounded Semantic Chunking:** Documents are chunked into contiguous windows of 800 to 1,000 characters with 150-character semantic overlaps. Chunks are strictly constrained within section boundaries (e.g. Introduction, Methodology, Experiments) and tagged with immutable metadata including physical page numbers.
3. **Dual-Stage Dense & Re-ranking Retrieval:** Embeddings are generated using BAAI/bge-small-en-v1.5 dense representations stored in ChromaDB. Queries undergo BM25 lexical expansion followed by dense cosine similarity matching and section-priority boosting.

3. Experiments and Datasets

To rigorously benchmark DeepScale, we evaluated the framework on four standardized research datasets:

- PubMedQA: A biomedical question-answering dataset containing 211.3k research abstracts labeled with reasoning steps.
- BioASQ 10b: An expert-curated biomedical benchmark testing complex query comprehension and multi-hop fact retrieval.
- ArXiv-CS: A corpus of 15,000 computer science preprints spanning machine learning, distributed systems, and computer vision.
- SCIDOCS: A multi-task scientific benchmark consisting of 1,000 query papers evaluated across citation recommendation and classification.

In total, over 35,000 experimental query-answer pairs were evaluated using automated LLM-as-a-Judge and human expert review.

4. Results and Major Findings

Our extensive experimental results yielded three major empirical findings:

1. Superior Factual Grounding: DeepScale achieved a factual consistency score of 94.2% on PubMedQA and 91.8% on ArXiv-CS, outperforming standard naive RAG baselines by 23.6 percentage points.
2. Significant Improvement in Generation Quality: Grounded generation with section-bounded chunks yielded an 18.4% relative improvement in ROUGE-L and a 14.1% increase in BERTScore (F1) over non-chunked context injection.
3. High Source Citation Precision: The automated citation alignment module achieved 96.5% page-level precision. In 965 out of 1,000 sampled test citations, the cited page and paragraph directly supported the synthesized claim without distortion.

5. Discussion and Limitations

Despite promising empirical gains, our study is subject to several important limitations:

1. **Latency Overhead on Very Large Documents:** Parsing and embedding massive monographs (e.g. over 150 pages) creates a noticeable latency bottleneck during initial ingestion, averaging 14.2 seconds per 100 pages on standard CPU hardware.
2. **Dependence on Clean OCR Quality:** When applied to scanned archival manuscripts with degraded optical character recognition (OCR) or complex non-standard font ligatures, text extraction accuracy dropped by 12.3%, leading to occasional chunk boundary misalignments.
3. **Complex Multi-Modal Elements:** DeepScale currently prioritizes textual and tabular sections; equations formatted in bitmap raster images and embedded vector graphics diagrams are not yet converted into semantic embeddings, potentially omitting mathematical proof details.
4. **High Memory Consumption for Extended Contexts:** Maintaining dense vector caches for multi-document comparisons can require significant RAM when scaling to thousands of concurrent papers.

6. Conclusion

We presented DeepScale, a grounded retrieval-augmented generation system designed specifically for research papers. By pairing layout-aware multi-column ingestion with section-aware chunking and verified page citations, DeepScale demonstrates that hallucination in scientific question answering can be dramatically curtailed. Future work will investigate end-to-end multi-modal chart and equation grounding.

7. References

- [1] Lewis et al., 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks', NeurIPS 2020.
- [2] Jin et al., 'PubMedQA: A Dataset for Biomedical Research Question Answering', EMNLP 2019.
- [3] Cohan et al., 'SPECTER: Document-level Representation Learning using Citation-informed Transformers', ACL 2020.